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The Art of Managing Flaky Tests at Scale

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***Abstract*—Test flakiness, characterized by non-deterministic test outcomes despite unchanged code, poses a significant challenge to continuous integration system efficiency and developer productivity. We comprehensively study a year-long initiative to mitigate test flakiness within a large-scale engineering organization. We adjust test failure sensitivity thresholds to account for alert fatigue and resolution quality, refocus data visualization for prioritization and pattern analysis, and introduce incentives to improve the rate of solutions contributing to test stabilization. Key findings reveal the importance of balancing technical efficiency gains with organizational dynamics.**

Keywords—continuous integration, testing, flaky, regression, software, automation

Test flakiness refers to non-deterministic behavior, where consecutive test executions yield different results despite any modifications to the code. A high rate of test flakiness increases developer toil, inflates spending, and reduces overall continuous integration ("CI") system efficiency. Maintaining stable CI system testing pipelines and high coverage is not just a technical issue but an art requiring tooling, process refinement, and cross-functional collaboration. A year-long initiative aimed at mitigating test flakiness within a large-scale engineering organization reveals:

1. The threshold at which action items get raised impacts desired outcomes.
2. Transparent and broadly accessed metrics across teams and systems build trust.
3. Aggregate evaluation of the quality of applied solutions uncovers systematic problems.

An internally developed system parses test result data and flags flaky tests when they have exceeded failure thresholds, a configurable setting. Action items must be addressed within one business day, as all required tests must pass before a change can be merged. Engineering teams operating within a single, monolithic repository allocate how much time teams spend balancing technical debt, including fixing flaky tests, with ongoing feature work. Over time, disabled test counts increased without a clear strategy to combat the cause and how to backtrack from this negative outcome. One impactful adjustment was **rightsizing flaky test alerting** to improve the proportion of test flakiness fixes applied within established deadlines. Fixes implemented on initial investigation across teams in aggregate nearly doubled - from 23.9% to 44.8% - by fine-tuning flaky test thresholds.

We measured how teams dealt with flaky test alerts and their quality of response when alerting thresholds were changed throughout a year of continuous experimentation by tracking the frequency at which a test fix was promptly applied. Our findings revealed that overly aggressive alerting thresholds led to a counterproductive "resolve now, investigate later" mentality, meaning that upon initial review, tests were temporarily disabled rather than promptly stabilized. Conversely, overly lenient thresholds enabled unstable tests to linger, resulting in increased flaky test rerun rates and decreased CI system efficiency.

A pivotal realization in our journey was that **alert fatigue significantly impacts long-term process adherence**. Teams have a finite capacity for addressing technical debt, and excessive alerts can lead to desensitization. Carefully adjusting alerting criteria increased the rate at which flaky tests were fixed as part of the investigation compared to our baseline, where most flaky test resolutions were disabled and left for future follow-up. We increased the percentage of promptly fixed flaky tests by understanding that teams can only handle a given amount of workload until undesirable behaviors emerge, namely, leaving test stability investigations for a later date. This adjustment improved flaky test resolution rates and reduced instances where teams resorted to disabling tests until the entire investigation was completed in the future.

Important metadata associated with flaky and disabled tests was already within our CI system but insufficiently formatted to answer pressing questions and establish patterns over time. Data visualization was reimagined to align with organizational hierarchy, and additional test metadata was added to support a cross-functional understanding of the state of our tests. The importance of data presentation aligning with the organization hierarchy was related to the developer's willingness to improve test stability rates. Existing data was not conveniently accessible nor coherently aggregated, leading to an imbalance of insight across role seniority and multiple teams. Data did not directly surface in the user interface of the CI system, and teams did not have the insight to query data sources directly. Surfacing additional metadata about the type of test and the modular ownership helped teams visualize how they stack up against one another in a widely accessible company resource. A unified dashboard enabled granular and aggregate analysis, simplifying the detection of patterns across the organization. Effectively visualizing patterns allowed us to focus on whether certain test types needed particular attention and enabled the establishment of performance trends across teams. Insufficient data

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granularity led to mis-scoped solutions, where a focused solution is implemented on every occurrence of a generalized problem.

To further support root cause analysis, we introduced a cross-disciplinary CI system incident calendar to capture breakages caused by external changes and system-wide intermittent instability. Increasing the accessibility of incident reporting about changes applied to upstream systems expedited root cause analysis, improving overall CI incident resolution time. Increased contextual information enabled teams to understand the entire system deeper, pinpoint systemic issues rather than treating each flaky test in isolation, and foster productive conversations to improve developer effectiveness further.

Investments in cultivating organization alignment and implementing diverse educational initiatives built a stronger testing ownership culture, including:

- Introducing **raffle-based rewards and shout-outs for engineers** contributing to proactive fixes in highly visible channels.
- Highlighting test stabilization patterns, best practices, and success stories with a **routine newsletter**.
- Developing **video tutorial content** on all aspects of understanding how tests run within the CI system.
- Scheduling **recurring office hours** to elevate conversations and introduce collaboration opportunities.

These initiatives **shifted the mindset around flaky tests from a noisy burden to collective responsibility**. Teaching about how the CI system operated, how to investigate common causes for flaky tests, and creating runbooks to improve test stability across various scenarios provided teams with confidence that improving test stability is a worthy investment.

Our experience underscores several key takeaways that engineering teams can apply:

1. Balance alerting thresholds into process design—technical efficiency and overly strict detection criteria lead to undesired outcomes in sociotechnical systems.
2. Improve data presentation—granular, well-structured visualization improves adoption and prioritization.
3. Invest in training and knowledge sharing—education drives cultural shifts that sustain long-term improvements.

However, the journey does not end here—flakiness will always be a moving target in evolving systems. No single strategy provides a "free lunch" in managing flaky tests. It requires organizational alignment and a commitment to continuous improvement. By reviewing test flakiness management processes, integrating flexible alerting, enhancing data visualization, and upskilling mixed with gamification, we have laid a foundation to tackle test flakiness at scale.

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